

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1101

G.M. Shrum 138 kV
November 03, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1101. The Interconnection Customer proposes a 246.6 MW natural gas combined cycle facility interconnecting at the G.M. Shrum 138 kV substation in the Kentucky area. The request was received on November 09, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 300 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$19,022,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Mcloughlin-Stateline 138kV
Contingency: Tristate-Hubbardstown 345kV
Loading: 130.1% (Rating: 322 MVA)
DFAX: 6.0%
Pre-project loading: 125.5%
2. Facility: Valemount Substation-Tristate 345kV
Contingency: Gavin Power Llc-Sporn 230kV
Loading: 123.8% (Rating: 1,168 MVA)
DFAX: 10.9%
Pre-project loading: 121.5%
3. Facility: Hyampom-Siggen Substation 138kV line
Contingency: Hubbardtown-Tristate 345kV
Loading: 122.2% (Rating: 395 MVA)
DFAX: 5.1%
Pre-project loading: 119.0%
4. Facility: G.M. Shrum-Palm 138kV
Contingency: Saf Hydroelectric Llc-G.M. Shrum 138kV
Loading: 117.2% (Rating: 192 MVA)
DFAX: 100.0%
Pre-project loading: 10.9%
5. Facility: Hyampom-Hubbardstown Substation 138kV
Contingency: Hubbardstown-Tristate Substation 345kV
Loading: 113.4% (Rating: 484 MVA)
DFAX: 5.5%
Pre-project loading: 110.5%

6. Facility: Valemount-Pence 345kV line
Contingency: Gavin Power Llc-Sporn 230kV
Loading: 110.9% (Rating: 1,069 MVA)
DFAX: 10.4%
Pre-project loading: 108.5%
7. Facility: Palm-Worthington Primary 138kV L
Contingency: Saf Hydroelectric Llc-G.M. Shrum Substation 138kV
Loading: 110.1% (Rating: 188 MVA)
DFAX: 100.0%
Pre-project loading: 21.1%
8. Facility: Mcloughlin-Hartland 138kV
Contingency: Hubbardstown-Tristate 345kV line
Loading: 107.3% (Rating: 408 MVA)
DFAX: 6.0%
Pre-project loading: 103.6%
9. Facility: Saf Hydroelectric Llc-G.M. Shrum 138kV
Contingency: Palm-G.M. Shrum 138kV
Loading: 105.7% (Rating: 213 MVA)
DFAX: 100.0%
Pre-project loading: 9.9%
10. Facility: Hanging Rock Energy Facility-Hanging Rock 138kV
Contingency: Hartland 138/1kV Autotransformer
Loading: 100.7% (Rating: 187 MVA)
DFAX: 7.0%
Pre-project loading: 91.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mcloughlin-Stateline 138kV	Rebuild line with higher-capacity conductor	\$3,403,000	22% (\$745,000)
Tristate-Valemount 345kV	Rebuild line with higher-capacity conductor	\$73,754,000	3% (\$2,542,000)
Hyampom-Siggen Substation 138kV line	Rebuild line with higher-capacity conductor	\$5,718,000	10% (\$552,000)
G.M. Shrum-Palm 138kV	Rebuild line with higher-capacity conductor	\$3,035,000	100% (\$3,035,000)
Hyampom-Hubbardstown Substation 138kV	Rebuild line with higher-capacity conductor	\$591,000	9% (\$54,000)
Valemount-Pence 345kV line	Rebuild line with higher-capacity conductor	\$38,216,000	4% (\$1,363,000)
Worthington Primary-Palm 138kV	Rebuild line with higher-capacity conductor	\$7,297,000	100% (\$7,297,000)
Mcloughlin-Hartland 138kV	Reconductor line	\$1,279,000	22% (\$280,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Saf Hydroelectric Llc-G.M. Hrum 138kV Line	Reconductor line	\$2,563,000	100% (\$2,563,000)
Hanging Rock Substation-Hanging Rock Energy Facility 138kV	Reconductor line	\$591,000	100% (\$591,000)
	Total allocated to project		\$19,022,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.